

R22

Code No: 181AB

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD

B. Tech I Year I Semester Examinations, September - 2023

BASIC ELECTRICAL ENGINEERING

(Common to CSE, IT, CSIT, CE(SE), CSE(CS), CSE(DS), CSD)

Time: 3 Hours

Max. Marks: 60

Note: This question paper contains two parts A and B.

i) **Part- A** for 10 marks, ii) **Part - B** for 50 marks.

- Part-A is a compulsory question which consists of ten sub-questions from all units carrying equal marks.
- Part-B consists of **ten questions** (numbered from 2 to 11) **carrying 10 marks each**. From each unit, there are two questions and the student should answer one of them. Hence, the student should answer five questions from Part-B.

PART- A

(10 Marks)

- 1.a) What are the limitations of Ohm's law? [1]
- b) State Norton's theorem. [1]
- c) Write any two advantages and disadvantages of 3-phase system. [1]
- d) Define power factor of an AC circuit. [1]
- e) Write the applications of auto-transformer. [1]
- f) Define the regulation of a transformers. [1]
- g) Why the core of the machine is laminated? [1]
- h) Why single-phase induction motor is not self-starting? [1]
- i) What is the function of MCCB? [1]
- j) What are the characteristics of batteries for longer life? [1]

PART - B

(50 Marks)

- 2.a) What is difference between a current source and a voltage source? Explain about ideal and practical voltage and current sources.
- b) A series R-C circuit consists of a resistor of $10\ \Omega$ and capacitor of $0.1\ \text{F}$ with a constant voltage of $20\ \text{V}$, is applied to the circuit at $t=0$. Obtain the current equation. Determine voltage across resistor and capacitor. [5+5]

OR

- 3.a) Obtain the Transient Response of series R-L circuit with DC excitation.
- b) Three resistances $2\ \text{ohms}$, $4\ \text{ohms}$ and $6\ \text{ohms}$ are connected in series across $24\ \text{V}$ supply. Find the voltages across three resistors and current through each resistor. Also calculate the total power consumed in the circuit. [5+5]

- 4.a) Derive expression of resonance frequency for series RLC circuit.
b) Derive the relation between line current and phase current in case of three phase delta connected balanced load. [6+4]

OR

- 5.a) A voltage of 125 V, 50 Hz is applied across a non-inductive resistor connected in series with a capacitor. The current in the circuit is 2.2 A. The power loss in the resistor is 96.8 W and that in the capacitor is negligible. Calculate the resistance and the capacitance.
b) With the help of phasor diagram, show that the current drawn by the R-L series circuit lags the applied voltage by an angle θ with respect to voltage. [5+5]

- 6.a) Draw and explain the equivalent circuit of a single-phase transformer.
b) The efficiency of a 10 kVA, 2000/400 V single phase transformer at unity power factor is 97% at rated load and at half rated load. Determine the transformer core losses and copper losses. [5+5]

OR

- 7.a) Explain briefly the principle of working of a transformer and show the voltage ratio of the primary and secondary windings is same as their turn's ratio.
b) Find the efficiency of a 150 kVA transformer at 25% of its full load at 0.8 power factor lagging if copper losses are 1600 W at full load and iron losses are 1400 W. [5+5]

- 8.a) With the help of a neat sketch, explain the construction and working principle of a DC machine.
b) The frequency of the e.m.f. in the stator of a 4-pole induction motor is 50 Hz, and that in the rotor is 2 Hz. What is the slip and at what speed is the motor running? [5+5]

OR

- 9.a) Explain the concept behind the generation of rotating field in a three-phase induction motor.
b) Explain the construction and working of a three-phase induction motor. [5+5]

- 10.a) Explain the working principle of SFU and MCB.
b) What are the drawbacks of low power factor? Describe how it is improved? [5+5]

OR

- 11.a) Explain the components of LT switch gear in detail.
b) Briefly explain following: (i) Need of Earthing (ii) Battery backup. [5+5]

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Code No: 181AB**JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD****B. Tech I Year I Semester Examinations, March/April - 2023****BASIC ELECTRICAL ENGINEERING****(Common to CSE, IT, CSIT, CE(SE), CSE(CS), CSE(DS), CSD)****Time: 3 Hours****Max. Marks: 60****Note:** This question paper contains two parts A and B.i) **Part- A** for 10 marks, ii) **Part - B** for 50 marks.

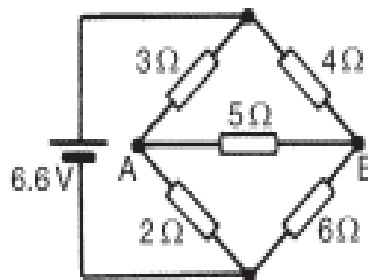
- Part-A is a compulsory question which consists of ten sub-questions from all units carrying equal marks.
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PART- A**(10 Marks)**

- 1.a) What is ideal voltage source? [1]
 - b) Define KVL. [1]
 - c) What is active power? [1]
 - d) Define average value of sinusoidal quantity? [1]
 - e) What is a step up Transformer? [1]
 - f) Define regulation of a transformer. [1]
 - g) What type of material is used for brushes of a d.c machine? [1]
 - h) How many windings are present at the time of starting a single phase Induction motor? [1]
- i) What is the full form of M.C.C.B? [1]
 - j) What is the energy consumed by 1000 watts heater in 3 hours? [1]

PART-B**(50 Marks)**

- 2.a) State and explain Thevenin's theorem.
- b) For the bridge network shown in figure below, find the current in the 5 resistor, and its direction, by using Thevenin's theorem. [5+5]

**OR**

- 3.a) Derive an expression for transient current in R-L series circuit excited by a d.c source.
 b) Explain the V-I relation of R, L and C elements. [6+4]

4. The following table gives the corresponding values of current and time for a half cycle of alternating current.

Time t (msec)	0	0.5	1.0	1.5	2.0	2.5	3.0	3.5	4.0	4.5	5.0
Current i (A)	0	7	14	23	40	56	68	76	60	5	0

Assuming the negative half cycle is identical in shape to the positive half cycle, plot the waveform and find (a) the frequency of the supply, (b) the instantaneous values of current after 1.25 ms and 3.8 ms, (c) the peak or maximum value, (d) the mean or average value, and (e) the r.m.s value of the waveform. [10]

OR

- 5.a) Derive an expression for resonance frequency of R-L-C series circuit connected to an alternating supply of variable frequency.
 b) A coil has an inductance of 40 mH and negligible resistance. Calculate its inductive reactance and the resulting current if connected to a 240 V, 50 Hz supply. [6+4]
- 6.a) Derive an e.m.f equation of a single phase transformer.
 b) A 4500 V/225 V, 50 Hz single-phase transformer is to have an approximate e.m.f. per turn of 15 V and operate with a maximum flux of 1.4 T. Calculate (i) the number of primary and secondary turns and (ii) the cross-sectional area of the core. [6+4]

OR

- 7.a) What is an Auto transformer? Give its advantages and state its applications.
 b) What are the advantages of 3-phase transformers? Draw various schemes of 3-phase transformer connections. [5+5]
- 8.a) Explain the constructional details of a d.c machine. Explain the working of a d.c generator.
 b) Draw the various characteristics of a d.c shunt motor. [6+4]

OR

- 9.a) Explain the working of a 3-phase Induction motor with a neat sketch.
 b) The stator of a 3-phase, 4-pole induction motor is connected to a 50 Hz supply. The rotor runs at 1455 rev/min at full load. Determine (i) the synchronous speed and (ii) the slip at full load. [6+4]
- 10.a) Explain the working of (i) SFU (Switch Fuse Unit) (ii) ELCB.
 b) What are the advantages of improving the power factor of a system? [6+4]

OR

- 11.a) What are the differences between primary cells and secondary batteries?
 b) Explain different types of wires used in domestic as well as commercial buildings. [4+6]

Code No: 181AB**JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD****B. Tech I Year I Semester Examinations, January/February - 2024****BASIC ELECTRICAL ENGINEERING****(Common to CSE, IT, CSIT, CE(SE), CSE(CS), CSE(DS), CSD)****Time: 3 Hours****Max. Marks: 60****Note:** This question paper contains two parts A and B.i) **Part- A** for 10 marks, ii) **Part - B** for 50 marks.

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PART- A**(10 Marks)**

- 1.a) Three capacitors of $4\ \mu\text{F}$, $4\ \mu\text{F}$ and $8\ \mu\text{F}$ are connected in parallel. What is its equivalent capacitance? [1]
- b) How Thevenin's voltage source and Thevenin's resistance are connected? [1]
- c) Define the peak factor. [1]
- d) What is the current and voltage relationship in a pure inductor? [1]
- e) What is the main purpose of using core in a transformer? [1]
- f) What is an autotransformer? [1]
- g) What is the basis for classification of D.C. Generators? [1]
- h) What is the equation of synchronous speed in a three-phase induction motor? [1]
- i) What is the primary objective of providing the circuit breaker in power system? [1]
- j) What are the types of circuit breakers are available in the market in the market for low voltage applications. [1]

PART - B**(50 Marks)**

- 2.a) State and explain KVL and KCL.
- b) Three resistors R_1 , R_2 and R_3 are connected in series across a constant voltage V . Voltage across R_1 is 20V. Power consumed by R_2 is 25W and $R_3 = 2\ \text{ohms}$. Find the voltage if the current is 5A. [5+5]

OR

- 3.a) State and explain the superposition theorem.
- b) A $500\ \mu\text{F}$ capacitor is charged to 20 V DC. It is suddenly connected to a resistance of $500\ \text{k}\Omega$. Find (i) voltage across capacitor at $t = 0$, (ii) current through capacitor at any time and (iii) voltage across capacitor at $t = \infty$. [5+5]

- 4.a) Define i) Instantaneous value, ii) Time period, iii) Frequency, iv) Cycle and v) Amplitude.
b) A sinusoidal voltage $v = 50 \sin \omega t$ is applied to a series RL circuit. The current in the Circuit is given by $i = 25 \sin (\omega t - 53^\circ)$. Determine (i) apparent power, (ii) Average power and (iii) power factor. [5+5]

OR

- 5.a) Derive the relation between line and phase quantities of voltages and currents for a delta connected system.
b) A coil has a resistance of 20 ohms and inductance of 80 mH and is connected in series with a 100 pF capacitor. Determine the circuit impedance at resonance and also find the resonant frequency? [5+5]
- 6.a) Explain different losses in a single-phase transformer.
b) A 220/110V, 50Hz, 1.5kVA, has primary and secondary winding resistance of 1Ω and 2Ω , and reactance of 3Ω and 5Ω respectively find the total equivalent resistance equivalent reactance and equivalent impedance referred by primary and secondary. [5+5]

OR

- 7.a) Explain the operation of auto transformer.
b) Describe the three-phase transformer connections. [5+5]
- 8.a) Explain the performance characteristics of dc shunt machine.
b) Discuss the construction and working of a three-phase induction motor. [5+5]

OR

- 9.a) Explain why single phase induction motor is not self-starting.
b) Describe the working principle of synchronous generator. [5+5]
- 10.a) Discuss the high rupturing capacity fuse with neat diagram. List out its merits and demerits.
b) Explain the ELCB and MCCB protection used in L.T switch gear. [5+5]

OR

- 11.a) What are the criterion considered for classified the wires and cables.
b) Describe the Nickel cadmium batteries. List out its merits and demerits. [5+5]

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